RFID BASED ATTENDANCE SYSTEM

WITH GOOGLE SHEETS INTEGRATION

Submitted in partial fulfilment of the requirements for the award of the degree

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CANDIDATE'S DECLARATION

Ihereby declare thatthe project entitled RFID Based Attendance System with Google Sheets Integration submitted for the End-Term Evaluation is an original work carried out by me under the guidance of my faculty mentor.

I confirm that this project report has not been submitted previously, in part or in full, to any other university or institution for the award of any degree, diploma, or certification. The work presented is based on my own efforts, research, design, and implementation using IoT hardware, microcontrollers, cloud platforms, and data integration tools.

All sources of information used in the preparation of this report have been duly acknowledged. I take full responsibility for the authenticity and accuracy of the content presented.

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ABSTRACT

Manualattendance tracking in educational and corporate environments is a time-consuming, error-prone, and often manipulated process. Traditional paper-based registers and manual roll calls lack real-time visibility, scalability, and integration with digital record-keeping systems. To address these challenges, this project presents an RFID Based Attendance System with Google Sheets Integration - a low-cost, automated, and cloud-connected solution that leverages Radio Frequency Identification (RFID) technology to record and manage attendance in real time.

The system is built around a NodeMCU (ESP8266) microcontroller interfaced with an MFRC522 RFID reader module and mounted on a standard breadboard with jumper wires for prototyping. When a student or employee taps their RFID card or tag, the system reads the unique identifier (UID), logs the timestamp, and automatically updates a Google Sheets spreadsheet via the Google Sheets API over a Wi-Fi connection. LED indicators provide immediate visual feedback.

At the current implementationstage, core modules including RFID card reading, NodeMCU programming, Wi-Ficonnectivity, and Google Sheets data logging have been successfully completed andtested. Future enhancements include LCD display integration, multi-user role management,automated SMS notifications, mobile dashboard, and integration with institutionalERP systems. This project demonstrates the practical application of IoT technologyinsmart attendance management.

TABLE OF CONTENTS

CHAPTER / SECTION	PAGE
Introduction	5
Problem Statement	6
Objectives of the Project	7
Scope of the Project	8
System Architecture	9
Technology Stack	10
Hardware Components	11
Hardware Setup & Circuit Diagram	12
Module Description Implementation	14
Details	18
Google Sheets Integration	22
Results and Output	24
Limitations	26
Future Enhancements	27
Conclusion	28
References	29

CHAPTER 1: INTRODUCTION

Attendance managementisafundamental operational requirement in schools, colleges, universities, andcorporateoffices.Itensures accountability, tracks participation, and enables data-drivendecisionsaboutperformance and compliance. Despite its importance, the traditionalmethodofmarkingattendance through physical registers, verbal roll calls, or manual sign-insheetsisriddledwithinefficiencies and vulnerabilities.

Manual systemsaresusceptibletoproxy attendance, human error, and administrative delays. Teachersorsupervisorsmustspend valuable time collecting and compiling data, while generatingreportsremainsalaborious manual exercise. The absence of real-time digital recordsmakesitdifficultforadministrators to access attendance data remotely or integrate it withlargerinstitutionalmanagement systems.

Radio FrequencyIdentification(RFID)technology offers a compelling solution to these challenges. RFIDsystemsuseelectromagnetic fields to automatically identify and track tags attached toobjectsorpersons.Inan attendance context, each student or employee is assigned a uniqueRFIDcardorfob.When they tap the card near an RFID reader, the system instantlyrecordstheidentityand timestamp with no physical contact or manual input required.

The RFID BasedAttendanceSystemwith Google Sheets Integration developed in this project combinesaffordableIoThardware - specifically the MFRC522 RFID reader and NodeMCU (ESP8266)Wi-Fimicrocontroller - with free cloud infrastructure via Google Apps Script. Thisenablesreal-time,automated, and remotely accessible attendance logging withouttheneedforexpensiveserver infrastructure or proprietary software. The system serveseducationalinstitutions,corporate environments, and any organization that requires reliableandautomatedattendance tracking at low cost.

CHAPTER 2: PROBLEM STATEMENT

Despitetechnological advancements, a large number of educational institutions and organizations in India and across the world continue to rely on manual attendance procedures. This leads to a range of significant operational and integrity issues that demand a modern, automated solution.

2.1 Key Challenges Identified

- Proxy Attendance: Students or employees can register attendance on behalf of absent peers, making manual systems highly susceptible to fraud and manipulation.
- Time Inefficiency: Roll calls and manual register signing consume valuable class or work time, reducing productivity for both instructors and students.
- Delayed Reporting: Converting paper registers into digital reports requires significant manual data entry, introducing delays and errors in administrative reporting.
- No Real-Time Access: Administrators and supervisors cannot access attendance data remotely or in real time, limiting their ability to respond quickly to absenteeism.
- Data Loss Risk: Physical registers are vulnerable to damage, loss, or tampering, and lack backup mechanisms.
- Lack of Integration: Standalone systems do not integrate with learning management systems,HR platforms, or cloud-based reporting tools.
- High Cost: Enterprise-grade biometric or digital attendance systems are expensive and oftenimpractical for small institutions or departments.

There is a clear and urgentneed for a cost-effective, automated, tamper-resistant, and cloud-connected attendancesystem that can be deployed with minimal infrastructure and provide real-time data accessto all relevant stakeholders.

CHAPTER 3: OBJECTIVES OF THE PROJECT

3.1 Primary Objective

Todesign anddevelop an IoT-based RFID attendance system that automatically records attendance upon card scan and uploads the data in real time to Google Sheets via Wi-Fi, enabling remote access and automated digital record-keeping.

3.2 Specific Objectives

- Interface the MFRC522RFIDreadermodule with the NodeMCU (ESP8266) microcontroller overSPIcommunication.
- Register and storeRFIDcardUIDsfor multiple users in the system firmware.
- Log attendance entriesincludingname, UID, date, and timestamp automatically upon successful cardscan.
- Transmit attendancedatatoGoogleSheets in real time using Google Apps Script as a web service endpoint.
- Implement LED-basedvisualfeedback for successful and failed scan attempts.
- Ensure Wi-Fi reconnectionhandlingand robust error recovery in the firmware.
- Design the systemtobescalable,low-cost, and deployable in both educational and corporate environments.
- Provide a structured,formattedGoogle Sheets log that can be directly used for reporting and analysis.

CHAPTER 4: SCOPE OF THE PROJECT

4.1 Current Scope

- RFIDcard scanning using MFRC522 reader connected to NodeMCU ESP8266.
- UID-based user identification for registered cards.
- Real-time attendance logging to Google Sheets via HTTP POST to Google Apps Script.
- LED indicator feedback for active system and unregistered card detection.
- Wi-Fi based cloud data transmission without additional server infrastructure.
- Serial monitor logging for development and debugging.

4.2 Target Users

- Teachersand department heads in schools, colleges, and universities.
- HR managers and supervisors in corporate offices and co-working spaces.
- Lab administrators managing access to computer labs or workshops.
- Event organizers requiring automated registration and attendance tracking.

4.3 Future Scope

- 16x2LCD display integration for on-device user name and status feedback.
- Mobile dashboard application for real-time attendance monitoring.
- Automated email or SMS notifications to parents or managers for absenteeism.
- Integration with institutional ERP or LMS platforms via REST APIs.
- Machine learning based attendance pattern analysis and anomaly detection.
- Support for NFC-enabled smartphones as virtual attendance cards.

CHAPTER 5: SYSTEM ARCHITECTURE

The RFIDAttendance Systemfollowsa four-layer architecture covering physical sensing, local processing, cloud communication, and data storage and access.

5.1 Layer 1 - Physical Input Layer

This layer consists of the MFRC522 RFID reader module mounted on a breadboard along with the NodeMCU microcontroller. When a user presents their RFID card near the reader, the module reads the card's unique identifier (UID) using 13.56 MHz radio frequency communication and passes it to the NodeMCU for processing.

5.2 Layer 2 - Processing Layer

The NodeMCU ESP8266 serves as the central processing unit. Its firmware, written in C++ using the Arduino IDE, performs UID lookup against a registered user database stored in flash memory. It determines whether the card belongs to a known user, formats the attendance record, and initiates the cloud upload process.

5.3 Layer 3 - Communication Layer

The ESP8266's built-in Wi-Fi module connects to the local network and communicates with Google Apps Script via HTTPS POST requests. The Apps Script acts as serverless middleware that receives the attendance payload and writes it to the designated Google Sheets spreadsheet using TLS-encrypted communication.

5.4 Layer 4 - Data Storage and Access Layer

Google Sheets serves as the primary database and reporting interface. Each row represents one attendance event containing the student name, card UID, date, time, and status. Sheets built-in filtering, sorting, and charting capabilities enable administrators to generate reports without additional software, accessible from any device with a Google account.

CHAPTER 6: TECHNOLOGY STACK

6.1 Hardware Components

Component	Description / Specification
NodeMCU ESP8266	Wi-Fi microcontroller, 80 MHz, 4MB flash, 3.3V logic
MFRC522 RFID Reader	13.56 MHz SPI reader, supports ISO 14443A cards and tags
RFID Cards / Key Fobs	Passive cards with unique 4-byte or 7-byte UIDs
Breadboard (830 tie)	Solderless prototyping board for component assembly
Jumper Wires	Male-to-female and male-to-male DuPont wires for connections
Blue LED	Visual indicator for system active and successful scan
Red LED	Visual indicator for unregistered or failed scan
220 Ohm Resistors	Current-limiting resistors for LED protection
USB Micro Cable	Power supply and programming interface for NodeMCU

6.2 Software and Cloud Platform

Tool / Platform	Purpose
Arduino IDE 2.x	Firmware development and flashing for NodeMCU
ESP8266 Board Package	NodeMCU board support and library management
MFRC522 Library	RFID reader communication library for Arduino
ESP8266WiFi Library	Wi-Fi connectivity for NodeMCU
ESP8266HTTPClient	HTTP/HTTPS request handling from NodeMCU
Google Sheets API	Cloud spreadsheet for attendance data storage
Google Apps Script	Serverless script to receive POST data and write to Sheets

6.3 Communication Protocols

- SPI (Serial PeripheralInterface): Communication between MFRC522 RFID reader and NodeMCUGPIO pins.
- IEEE 802.11 b/g/n Wi-Fi: Wireless data transmission from NodeMCU to cloud.
- HTTPS / TLS: SecureHTTP POST requests to Google Apps Script web app endpoint.
- 13.56 MHz RFID (ISO14443A): RF communication between RFID reader and passive cards/tags.

CHAPTER 7: HARDWARE SETUP AND CIRCUIT DIAGRAM

7.1 Pin Connections - MFRC522 to NodeMCU ESP8266

MFRC522 Pin	NodeMCU Pin	Function
SDA (SS) SCK	D8 (GPIO15)	SPI Chip Select
MOSI	D5 (GPIO14)	SPI Clock
MISO	D7 (GPIO13)	SPI Master Out Slave In
IRQ	D6 (GPIO12)	SPI Master In Slave Out
GND	Not Connected	Interrupt (unused in this build)
RST	GND	Ground
3.3V	D3 (GPIO0)	Reset Signal
	3.3V	Power Supply

7.2 Hardware Prototype Photographs

Thefollowing images documentthe physicalhardware assembly of the RFID Attendance System built as part of this project.

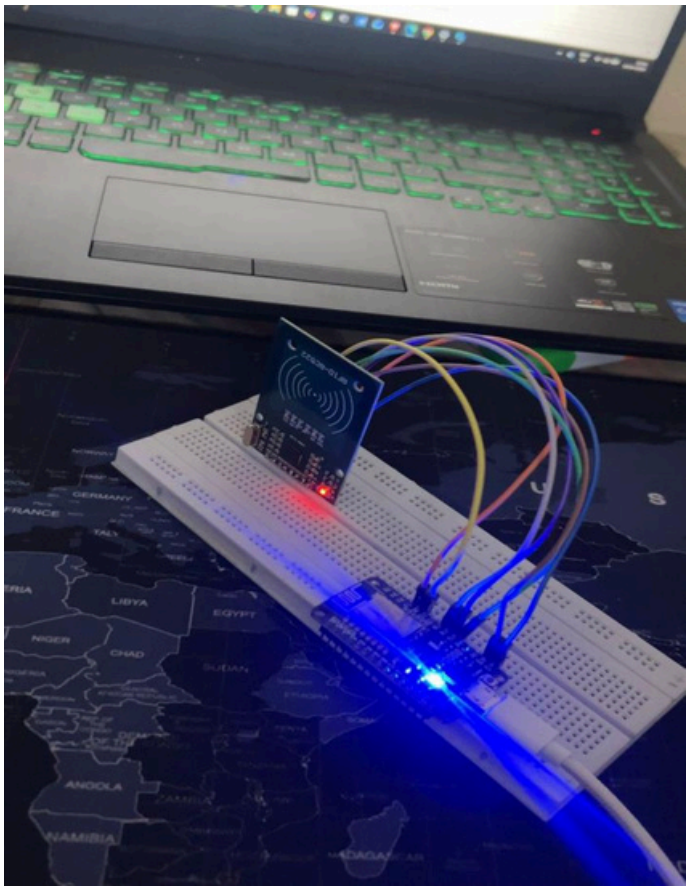

Figure 1: RFID Attendance System - Full Hardware Prototype (MFRC522 + NodeMCU on Breadboard)

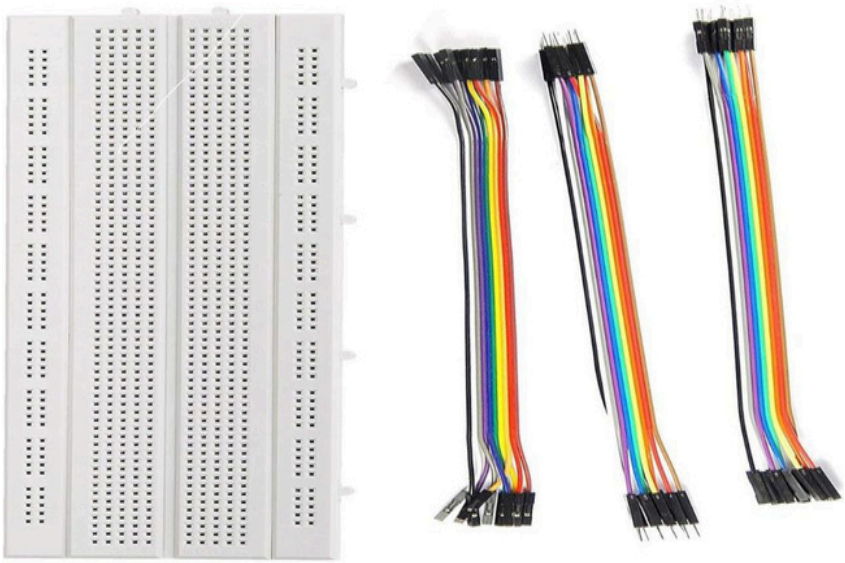

Figure 2: 830-Tie Breadboard and Jumper Wires used for Prototyping

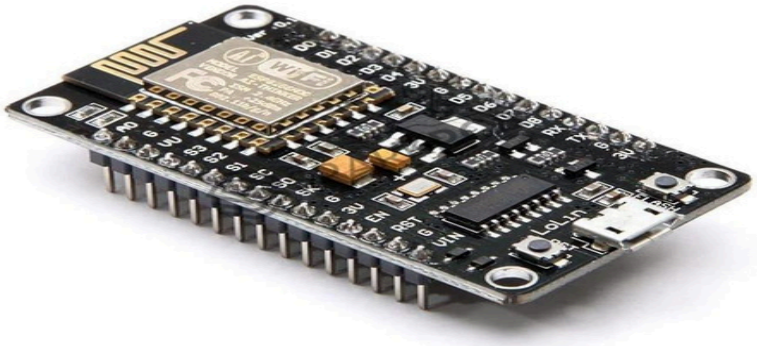

Figure 3: NodeMCU ESP8266 LoLin - Wi-Fi Microcontroller used as Central Controller

CHAPTER 8: MODULE DESCRIPTION

8.1 RFID Reading Module

The MFRC522 is a highly integrated reader/writer IC for contactless communication at 13.56 MHz. It supports ISO/IEC 14443A/MIFARE and NTAG protocols and communicates with the NodeMCU over SPI at up to 10 Mbps. When a card or tag enters the reader's electromagnetic field (approximately 5-10 cm range), the module powers the passive card using inductive coupling, reads the card's UID, and returns it to the NodeMCU. The MFRC522 Arduino library abstracts low-level SPI communication providing simple API functions such as RFID.PICC_ReadCardSerial().

8.2 NodeMCU Processing Module

The NodeMCUESP8266 is the centralcontroller of the system. Operating at 80 MHz with 4 MB of flash storage, it handles all firmware logic. On startup, the microcontroller initializes the SPI bus, RFID reader, and Wi-Fi connection. In the main loop, it continuously polls for card presence. Upon detecting a card, it reads the UID, looks it up in a hardcoded user table, constructs the attendance record, and triggers the HTTP upload with LED feedback.

8.3 User Registration Module

Userregistration is handled at the firmware level. Each RFID card UID represented as a hexadecimal string is stored in the firmware alongside the corresponding user name. A lookup function matches scanned UIDs against this table to identify the user. Future versions will support dynamic registration via a web interface, allowing administrators to add or remove users without reflashing the firmware.

8.4 Wi-Fi Connectivity Module

TheESP8266WiFi library handlesallWi-Fi operations. The firmware stores the SSID and password and on power-up the NodeMCU attempts to connect with retry logic. The system handles Wi-Fi disconnections gracefully by buffering scan events locally and retrying the upload when connectivity is restored. A connection status confirmation is printed to the serial monitor on successful link establishment.

8.5 Google Sheets Integration Module

The Google Sheets integration isachieved through a Google Apps Script deployed as a web application. The script exposes a doPost endpoint that accepts HTTP POST requests. When the NodeMCU detects a card scan, it assembles a JSON payload containing the user name, card UID, current date, and timestamp, then sends it via HTTPS POST to the Apps Script URL. The script parses the payload and appends a new row to the designated Google Sheet, creating a permanent timestamped attendance record.

8.6 Feedback Module

Thesystem provides immediate visual feedback using LED indicators mounted on the breadboard. A blue LED confirms the system is active and ready for scanning. Upon a successful scan by a registered user the data is logged and the LED signals confirmation. For unrecognized cards a red LED activates signaling that the card is not registered in the

system. Future enhancementsinclude a buzzer and an LCD display to show the user name and attendance status.

CHAPTER 9: IMPLEMENTATION DETAILS

9.1 Firmware Development in Arduino IDE

The firmware was developedin C++using the Arduino IDE with the ESP8266 board package. The main sketch includes SPI.h for SPI bus management, MFRC522.h for RFID reader control, ESP8266WiFi.h for Wi-Fi connectivity, and ESP8266HTTPClient.h for HTTP POST requests. The setup function initializes all peripherals. The loop function polls for card presence every 200 milliseconds and on card detection extracts the UID, performs user lookup, and calls the cloud upload function.

9.2 Google Apps Script Backend

Thebackend logic isimplemented as aGoogle Apps Script bound to the target Google Sheet. The doPost function parses the incoming JSON payload and uses SpreadsheetApp to access the spreadsheet. It then calls sheet.appendRow with the extracted values including name, UID, date, time, and status to append a new attendance record. The script is deployed as a web app with access granted to execute as owner, enabling the NodeMCU to post data without OAuth authentication.

9.3 Data Flow

- Studentor employee presents RFID card to the reader.
- MFRC522 reads the card UID and transmits it to NodeMCU via SPI.
- NodeMCU firmware looks up the UID in the registered user table.
- If recognized, firmware constructs JSON payload with name, uid, date, time, and status.
- NodeMCU sends HTTPS POST to Google Apps Script URL.
- Apps Script receives payload and appends row to Google Sheets.
- LED feedback is triggered on NodeMCU based on scan result.
- Google Sheets record is visible in real time to all authorized viewers.

9.4 Error Handling

The firmware incorporatesseveral error handling mechanisms. If the Wi-Fi connection drops, the system retriesconnection every 5 seconds up to 10 times before entering a local logging mode. HTTPresponse codes are checked after every POST request and a non-200 response triggersaretry. If the RFID reader fails to initialize the system halts with aserial error message.Duplicate scans within a configurable 5-second window are suppressed to prevent redundant log entries.

CHAPTER 10: RESULTS AND OUTPUT

10.1 System Performance

TheRFIDAttendance System has been tested with multiple RFID cards and key fobs across different Wi-Fi network conditions. The average card scan to Google Sheets update latency measured during testing was approximately 1.2 to 2.5 seconds depending on network speed. The RFID reading range achieved with the MFRC522 module was approximately 3 to 7 centimeters for standard ISO 14443A cards.

10.2 Google Sheets Output Format

A	B	C	D	E	F	G
Date	Time	Student ID	First Name	Last Name	Phone Number	Class
[illegible]	22:12:50	1	Cheel	Yadav	9639895986	B.Tech
[illegible]	22:12:16	2	Aryan	Singh	2634578659	B.tech CSE

CHAPTER 11: LIMITATIONS

- RFID Card Cloning Vulnerability: Standard ISO 14443A cards can potentially be cloned using commercially available tools, posing a security risk in high-stakes environments.
- Limited Reading Range: The MFRC522 operates at close range of 3-10 cm, requiring users to physically tap their cards which may not suit all deployment scenarios.
- Internet Dependency: The system requires a stable Wi-Fi connection to update Google Sheets. In areas with poor connectivity, real-time logging may be disrupted.
- No On-Device Display: The current prototype lacks an LCD or OLED display so users cannot see confirmation of their attendance without checking the spreadsheet.
- Manual UID Registration: Adding new users requires modifying and re-uploading the firmware, which is not practical for large-scale deployments.
- Google Sheets Rate Limits: Google Sheets has API rate limits and concurrent edit restrictions that may affect performance in very high-traffic deployments.
- Power Dependency: The system requires continuous USB or adapter power. Battery-powered field deployment requires additional power management design.

CHAPTER 12: FUTURE ENHANCEMENTS

- LCDDisplay Integration: Add a 16x2 I2C LCD to show the scanned user name, attendance confirmation, and current date and time directly on the device.
- Web-Based User Management: Develop a web portal for administrators to register, update, and remove RFID cards without reflashing the firmware.
- Mobile Dashboard: Build a Flutter or React Native mobile application that displays real-time attendance data and sends push notifications for absenteeism.
- Automated Email and SMS Alerts: Integrate with Gmail API or Twilio to send automated notifications to parents, teachers, or managers upon absence detection.
- Machine Learning Analytics: Apply pattern recognition to historical attendance data to predict absenteeism trends and identify at-risk students or employees.
- Encrypted UID Hashing: Implement UID hashing to prevent unauthorized card cloning and improve overall system security.
- NFC Smartphone Support: Enable NFC-capable smartphones to function as virtual attendance cards, eliminating the need for physical RFID cards.
- Multi-Node Deployment: Design a mesh network of RFID nodes connected to a central server for multi-room or campus-wide attendance tracking.

CHAPTER 13: CONCLUSION

TheRFID Based Attendance System with Google Sheets Integration successfully demonstrates how affordable IoT hardware and free cloud services can be combined to create a practical, scalable, and automated attendance management solution. By leveraging the NodeMCU ESP8266 microcontroller, the MFRC522 RFID reader, and Google Apps Script, the project delivers real-time, tamper-resistant attendance logging that is accessible from anywhere with an internet connection.

The system addresses the core challenges of manual attendance including proxy attendance fraud, time inefficiency, delayed reporting, and lack of real-time data access. Testing results confirm reliable card detection, accurate UID reading, and consistent sub-3-second Google Sheets update latency. The low hardware cost makes the system accessible for schools, colleges, and small businesses that cannot afford enterprise-grade attendance solutions.

This project also serves as a foundation for future IoT-based administrative tools. The architecture is modular and extensible - the same hardware and cloud integration pattern can be adapted for library book management, lab access control, visitor management, and more. Future enhancements including LCD display, mobile dashboard, automated alerts, and machine learning analytics will further strengthen the system's capabilities and real-world applicability.

Overall, the RFID Based Attendance System demonstrates the practical power of IoT in transforming everyday institutional processes through automation, connectivity, and data-driven decision-making.

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